

Project #3 Report by Tomas Salaz

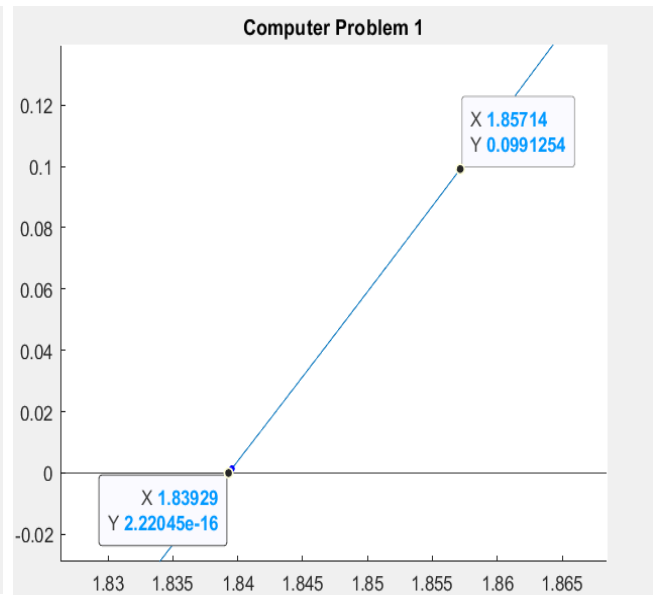
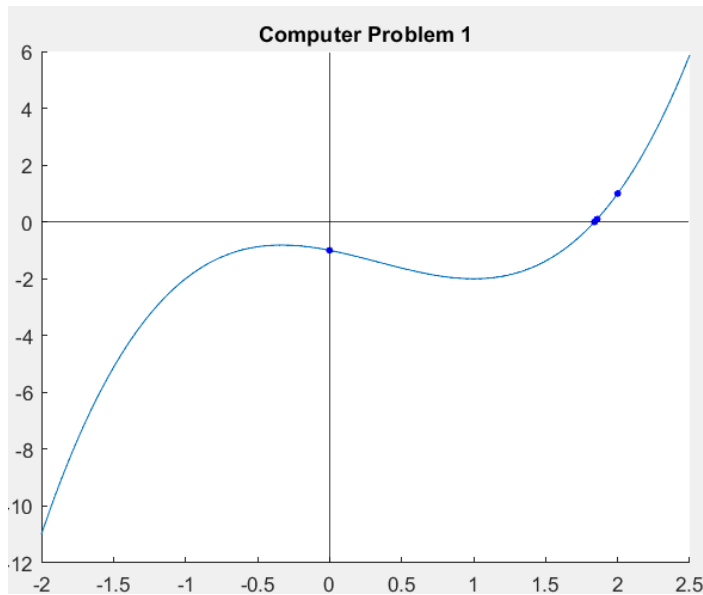
Written Problems 1-3, See .pdf in Canvas

Computer Problem 1) Approximate $x^3 = x^2 + x + 1$ within 10^{-8}

Summary:

According to the Plot and Console Output (see images below), the functions root is approximately at $x = 1.83928676$.

The plot shows the line for the function $f(x) = x^3 - x^2 - x - 1$ (in Blue) along with the points from Newtons Method. These points are hard to see so there is a zoomed in picture on the right, the attached image of the console also shows the array of values and the number of iterations it took to find the root.



```
>> newtMethod1
Converged to x = 1.83928676 in 6 iterations
2.00000000
1.85714286
1.83954451
1.83928681
1.83928676
1.83928676
```

Computer Problem 2) Find Positive Root of $2x(1+x^2)^{-1} = \arctan(x)$.

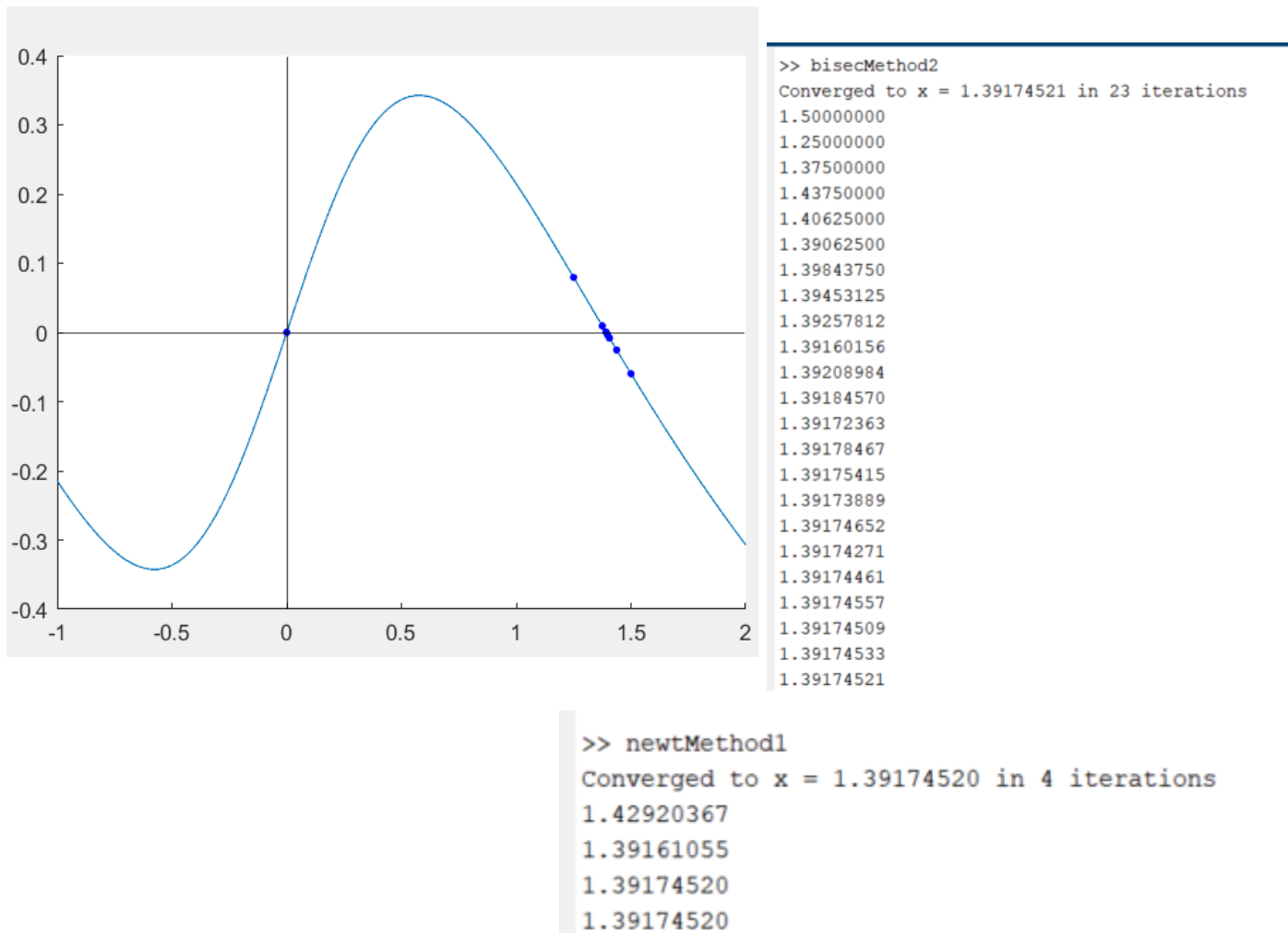
Summary:

The Plot and Console output approximate the function

$f(x) = \frac{2x}{(1+x^2)} - \arctan(x)$ has a root at $x = 1.39174521$.

The Bisection Method is much slower than Newtons Method, as this process took 23 Iterations versus 4 Iterations with $x_0 = 1$. Bisection is more foolproof however, since the function is even so the roots are the same on either side, so bisection on the interval $(0, \text{infinity})$ will work better than Newtons with a bad initial guess. So, there is a tradeoff of Efficiency for 'Accuracy' in some sense.

The picture below are the Plot of the function with the points as Newtons Method Converges and the console outputs for the Bisection Method Version and the Newtons Method Version with $x_0 = 1$.



Computer Problem 3) Solve the System using Newtons Method with $x_0 = 1$.

$$\begin{cases} x^3 - 2xy + y^7 - 4x^3y = 5 \\ y \sin(x) + 3x^2y + \tan(x) = 4 \end{cases}$$

Summary

Since the 2nd equation can be solved for y , I used Elimination, so I solved for y and plugged in that y value to the top equation. Then I used Newtons Method to solve they system, using only that equation, Y and the function $f(x)$ are as follows

$$y = \frac{4 - \tan(x)}{\sin x + 3x^2}$$

$$f(x) = -5 + x^3 - 2xy + y^7 - 4x^3$$

Using $f(x)$ and its derivative (using Finite Difference Approx*) along with the Newtons Method Algorithm, the x value that solved they system is $x = 13.88948331$ and it got there in 10 iterations.

*References

[Numerical Integration and Differentiation - MATLAB & Simulink \(mathworks.com\)](https://www.mathworks.com)

[6: Finite Difference Approximation - Mathematics LibreTexts](https://math.libretexts.org)

[Lecture 3 Part 2: Finite-Difference Approximations | Matrix Calculus for Machine Learning and Beyond | Mathematics | MIT OpenCourseWare](https://ocw.mit.edu/courses/mathematics/6-035-linear-algebra-and-differential-equations-i/)

[Finite Difference \(FD\) Approximation \(mit.edu\)](https://ocw.mit.edu/courses/mathematics/6-035-linear-algebra-and-differential-equations-i/)

Here is the Console output, Shows the Convergence and the amount of iterations for this version of Newtons Method.

```
Converged to 1.388948e+01.8 in 10 iterations
1.00000000
13.66591313
14.10538048
14.07772897
14.03270307
13.97256522
13.91757694
13.89271601
13.88952626
13.88948331
```